Github
Linkedin

Nikola Milosavljević

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Programming languages:
Proficient: Python | Rust
Familiar: Go | Java | SQL

Software Engineer specializing in distributed systems, scalability, and performance optimization. Strong foundation in theoretical computer science and algorithm design, with experience in information retrieval, execution runtimes, and storage infrastructure.

Experience

Site Reliability Engineer — **DFINITY Stiftung** — **Zürich** — **09/2022 - Present**

Observability & automation - platform powering engineers with insights into thousands of machines at low cost and latency.

- Designed and implemented a scalable service discovery module powering the company's **observability platform**, synchronizing global system state, integrating with **Prometheus** and **VictoriaMetrics** and generating dynamic observability configurations.
- Architected the company's end-to-end logging infrastructure; led migration from AWS to on-prem Kubernetes, reducing costs by **95%** and supporting **30M logs/hour**.
- Constructed an automated release qualification pipeline, executing pre/post-upgrade performance tests to quantify regressions and guide release decisions.

Opencloud - frontier sovereign serverless cloud platform for AI agents. [opencloud.org]

- Engineered core components of the **control plane**, allowing on-demand cloud provisioning and insights.
- Piloted **AI nodes**, allowing customers to keep ownership of their data while providing powerful features for their **cloud engines**.
- Prototyped an **SRE agent**, giving engine owners helpful insights and automatic detections of anomalies happening on their platform.

Software engineering [dfinity/ic] [dfinity/dre]

- Automated node migrations that are reducing swapping time from **days** to **seconds** while improving security and flexibility.
- Introduced a linear optimization tool for **distributed network design**, enabling constraint-based topology generation and capacity planning for target network states.

Outsourced Fullstack Software Engineer — **Schneider Electric** — **Novi Sad** — **05/2021 - 09/2022**

- Built a billing determinant module integrating IoT devices and handling **terabytes of data** with reliable large-scale processing.
- Developed smart meter orchestration system enabling users to remotely configure, monitor and diagnose devices usiong well known protocols, improving observability and operation control at scale.
- Led a company wide infrastructure provisioning refactor, introducing automation with terraform and ansible.

Teaching assistant — **University of Novi Sad** — **Novi Sad** — **09/2023 - 09/2024**

Microservice architecture

- Lead labs and assisted lectures for 300+ students across multiple university groups.
- Taught students concepts of gRPC, message queuing and saga patterns.

Introduction to cybersecurity

- Delivered introductory cybersecurity lectures to 150+ students, teaching core concepts.
- Held a hackaton, allowing students to learn and use penetration testing, basics of cryptography and public key infrastructure.

Education

MSc in Computer Science — **Unvieristy of Novi Sad - FTN** — **09/2022 - 12/2023**

- GPA: **9.75/10** | **Cybersecurity and payment systems section**
- Graduated in the top of the class (in top **1%**)

BSc in Computer Science — **Unvieristy of Novi Sad - FTN** — **09/2018 - 07/2022**

- GPA: **9.49/10** | **Computer software and automation section**
- Graduated in the top of the class (in top **1%**)

Honors

National Fellowship — **University of Novi Sad, 2018–2022**

- Serbia's national fellowship for academic merit for top 1% of students, awarded for four consecutive years.

National CyberHero hackaton — **University of Belgrade, 2021**

- Placed **5th**, solving security challenges across multiple domains.